Light - Reflection and Refraction

LIGHT - REFLECTION AND REFRACTION

NCERT Class 10 Science • Ch 9 | Hinglish Detailed Notes

NCERT Class 10 Science - Chapter 9 (Hinglish Detailed Notes)

COLOUR CODE: RED = Core Basics / extra help (Class 7-8-9 wali cheezein jo yaad honi chahiye).

COLOUR CODE: GREEN = Board exam me ye topics BAAR-BAAR aate hain - pakka yaad karo.

Baaki normal black text = main chapter content (detail me, simple bhasha me).

CORE BASICS - PEHLE YE PADHO (CLASS 7-8-9 KA FOUNDATION)

Tune 7-8-9 nahi padhi - koi tension nahi. Ye page is chapter ki base hai.
Ek baar achhe se padh le, fir poora chapter aasaani se samajh aayega.

1) LIGHT (prakaash) kya hai:
- Ek tarah ki energy jisse hume cheezein DIKHTI hain.
- Light hamesha SEEDHI (straight line) me chalti hai.

2) RAY aur BEAM:
- RAY (kiran) = light ke chalne ki ek seedhi patli line (arrow se dikhate).
- BEAM (puhnj) = bahut saari rays ka bundle/group.

3) LUMINOUS vs NON-LUMINOUS:
- LUMINOUS = khud roshni dete (Sun, bulb, jugnu, aag).
- NON-LUMINOUS = khud roshni nahi dete, doosri light se dikhte (chaand, mez, kitaab).

4) TRANSPARENT / TRANSLUCENT / OPAQUE:

- TRANSPARENT = light poori paar (saaf kaanch, paani) - aar-paar dikhta.
- TRANSLUCENT = light thodi paar (butter paper, dhundli kaanch) - dhundla dikhta.
- OPAQUE = light bilkul paar nahi (lakdi, deewar, lohaa).

5) REFLECTION (Class 7) kya hai:
- Light kisi chamakdaar surface (mirror) se TAKRA ke WAPAS aati hai = reflection.
- Isi se aaina (plane mirror) me apni image dikhti hai.

6) NORMAL, ANGLE OF INCIDENCE, ANGLE OF REFLECTION:

- NORMAL = surface ke point par 90 degree par khinchi gayi line (lambvat).
- ANGLE OF INCIDENCE (i) = incident (aane wali) ray aur normal ke beech ka angle.
- ANGLE OF REFLECTION (r) = reflected (laut-ne wali) ray aur normal ke beech ka angle.

7) REAL vs VIRTUAL IMAGE (bahut zaroori):
- REAL image = jahan rays sach-much milti hain; screen par PAKDI ja sakti; ulti (inverted).
- VIRTUAL image = jahan rays sach me nahi milti (bas aisa lagta); screen par NAHI aati; seedhi (erect).

8) CONCAVE vs CONVEX (intro):
- CONCAVE = andar dhansa hua (spoon ka andar wala part) - "converging".
- CONVEX = bahar ubhra hua (spoon ka peeche wala part) - "diverging".

9) CENTRE aur RADIUS ka idea:
- Spherical mirror/lens ek bade sphere (gend) ka tukda hota hai.
- Us sphere ka centre = centre of curvature (C); radius = radius of curvature (R).

10) UNITS:

- Length/distance: cm ya m (1 m = 100 cm).
- Lens ki POWER: DIOPTRE (D). Focal length (f) hamesha METRE me daalo for power.

11) REFRACTION ka matlab:

- Jab light ek medium se doosre (jaise hawa -> paani/kaanch) jaati hai to apna raasta MOD leti hai (bend ho jaati) - isko REFRACTION kehte.

1. REFLECTION OF LIGHT

LAWS OF REFLECTION (do niyam - definition zaroor yaad karo):

(i) Angle of incidence = Angle of reflection, yaani angle i = angle r.

(ii) Incident ray, reflected ray aur normal (point of incidence par) - teeno EK HI PLANE (same plane) me hote hain.

Ye dono niyam plane aur spherical (concave/convex) sab mirrors par lagte hain.

1.1 PLANE MIRROR SE IMAGE (aaina)

Plane mirror se bani image ke 5 properties (exam favourite):

(i) VIRTUAL aur ERECT (seedhi) hoti hai.

(ii) Object jitni BADI - SAME SIZE.

(iii) Mirror ke PEECHE banti, utni hi door jitna object aage hai.

(iv) LATERALLY INVERTED - left-right ulat jaata (ambulance ka ULTA likha word sheeshe me seedha dikhta).

2. SPHERICAL MIRRORS (CONCAVE AND CONVEX)

Spherical mirror = sphere ke tukde jaisa curved mirror. Do type:

- CONCAVE (converging): reflecting surface ANDAR ki taraf (dhansa hua).
- CONVEX (diverging): reflecting surface BAHAR ki taraf (ubhra hua).

2.1 ZAROORI TERMS (yaad rakho)

- POLE (P) = mirror ka centre point (beech).
- CENTRE OF CURVATURE (C) = us sphere ka centre jiska mirror tukda hai.
- RADIUS OF CURVATURE (R) = P se C tak ki doori (sphere ka radius).
- PRINCIPAL AXIS = P aur C ko jodne wali seedhi line.
- PRINCIPAL FOCUS (F) = principal axis ke paas wali parallel rays reflect hokar jis point par milti (concave) ya jahan se aati lagti (convex) hain.
- FOCAL LENGTH (f) = P se F tak ki doori.

BAHUT IMPORTANT RELATION: $f = R/2$ (focal length, radius of curvature ka aadha)

3. RULES FOR RAY DIAGRAMS (MIRRORS)

Image banane ke liye kisi bhi 2 rays kaafi hain. Standard rules:

RULE 1: Principal axis ke PARALLEL aane wali ray, reflect hokar -
- Concave: FOCUS (F) se hokar guzarti.
- Convex: F se aati hui lagti (peeche se).

RULE 2: F se hokar (ya F ki taraf) aane wali ray, reflect hokar principal axis ke PARALLEL ho jaati hai.

RULE 3: Centre of curvature (C) se hokar (ya C ki taraf) aane wali ray, usi raaste par WAPAS (retrace) chali jaati (kyunki wo normal par padti hai).

4. IMAGE FORMATION BY CONCAVE MIRROR

Object ki 6 positions ke liye image (ye TABLE exam me pakka - ratlo):

1) Object at infinity -> Image: at F; highly diminished (point); real, inverted
2) Object beyond C -> Image: between F and C; diminished; real, inverted
3) Object at C -> Image: at C; same size; real, inverted
4) Object between C and F -> Image: beyond C; enlarged; real, inverted
5) Object at F -> Image: at infinity; highly enlarged; real, inverted
6) Object between F and P -> Image: behind mirror; enlarged; VIRTUAL, erect

Yaad karne ka trick: object jaise-jaise C se F ki taraf aata, image badi hoti jaati. Sirf LAST case (F aur P ke beech) me image VIRTUAL + seedhi.

5. IMAGE FORMATION BY CONVEX MIRROR

Convex mirror me object kahin bhi rakho - image hamesha EK HI tarah:

- VIRTUAL, ERECT (seedhi), aur DIMINISHED (chhoti) hoti hai.
- Image hamesha mirror ke PEECHE, P aur F ke beech banti hai.

Do positions: (i) object at infinity -> image at F (point, behind);
(ii) object kahin bhi (P aur infinity ke beech) -> image P aur F ke beech.

5.1 CONVEX MIRROR KE USES

- Gaadi ka REAR-VIEW (side) mirror - kyunki erect + chhoti image deta aur bada area (wider field of view) dikhata (peeche ka zyada traffic dikhta).
- Shops/parking me SECURITY mirror.

6. USES OF SPHERICAL MIRRORS

CONCAVE MIRROR ke uses (yaad rakho):

- Torch, search-light, gaadi ki HEADLIGHT ke reflector (bulb F par rakh ke parallel strong beam milti).
- SHAVING mirror / makeup (chehra paas rakhne par bada seedha dikhta).
- DENTIST ke paas (daant ka bada image dekhne ko).
- SOLAR furnace / solar cooker (suraj ki garmi ek point par focus karne ko).

7. SIGN CONVENTION (NEW CARTESIAN)

Numerical solve karne se PEHLE sign convention samajhna zaroori:

- Saari doori POLE (P) [mirror] / OPTICAL CENTRE (O) [lens] se naapte hain.
- Object hamesha LEFT me rakhte; light left -> right jaati.
- Incident light ki direction me (right) doori = +ve; ulti taraf (left) = -ve.
- Isliye OBJECT DISTANCE (u) hamesha -ve (object left me).
- Principal axis ke UPAR height = +ve; NEECHE = -ve.

Isi se: Concave mirror ka f = -ve, Convex mirror ka f = +ve.
Convex lens ka f = +ve, Concave lens ka f = -ve.

8. MIRROR FORMULA AND MAGNIFICATION

MIRROR FORMULA: $1/v + 1/u = 1/f$
v = image distance, u = object distance, f = focal length (sign ke saath).

MAGNIFICATION (m): $m = -v/u = h'/h$
h' = image height, h = object height.

- m -ve -> image REAL aur INVERTED (ulti).
- m +ve -> image VIRTUAL aur ERECT (seedhi).
- |m| > 1 -> enlarged (badi); |m| < 1 -> diminished (chhoti); |m| = 1 -> same size.

9. REFRACTION OF LIGHT

REFRACTION = light jab ek paardarshi (transparent) medium se doosre me jaati hai to apni speed badalti aur isliye MUD (bend) jaati hai.

- Rarer -> Denser (hawa -> kaanch): light normal ki TARAF mudti.
- Denser -> Rarer (kaanch -> hawa): light normal se DOOR mudti.
- Normal ke saath aaye to bina mude seedha nikal jaati.

LAWS OF REFRACTION (do niyam):

(i) Incident ray, refracted ray aur normal - teeno SAME PLANE me hote.

(ii) SNELL'S LAW: kisi do media ke jode ke liye sin i / sin r constant rehta.
Yaani $n = \sin i / \sin r$ (= refractive index).

9.1 REFRACTIVE INDEX (n)

REFRACTIVE INDEX = ek medium me light kitni dheemi padti, uska maap.

$n = c/v$

c = light ki speed VACUUM/hawa me (3×10^8 m/s), v = us medium me speed.

- n jitna ZYADA, medium utna "optically DENSER" (light utni dheemi).
- OPTICALLY DENSER = zyada n (jaise kaanch); RARER = kam n (jaise hawa).
- (Optically denser ka density se direct lena-dena zaroori nahi.)

9.2 RECTANGULAR GLASS SLAB SE REFRACTION

- Light slab me ghuste waqt normal ki taraf mudti, nikalte waqt normal se door.
- EMERGENT ray, incident ray ke PARALLEL hoti hai (sirf thodi side khisak jaati).
- Is khiskaav ko LATERAL SHIFT (lateral displacement) kehte.

9.3 ROZ-MARRA UDAHARAN

- Paani me rakhi STICK/pencil tedhi (bent) dikhti - kyunki paani se aati light surface par refract hoti, isliye stick ka doobaa hissa uthaa/muda dikhta.
- Isi wajah se paani ka tank kam gehraa (shallow) dikhta, taare timtimaate hain.

10. REFRACTION THROUGH LENSES

LENS = do surfaces se bana paardarshi kaanch jo refraction se light ko mod/jod deta.
- CONVEX lens = beech me MOTA, kinaare patle -> CONVERGING (rays ko ek point par jodta).
- CONCAVE lens = beech me PATLA, kinaare mote -> DIVERGING (rays ko faila deta).

10.1 ZAROORI TERMS

- OPTICAL CENTRE (O) = lens ka beech ka point (ray bina mude nikal jaati).
- PRINCIPAL FOCUS (F) = axis ke parallel rays jis point par milti (convex) ya jahan se aati lagti (concave). Lens ke DONO taraf focus hota (F1, F2).
- FOCAL LENGTH (f) = O se F tak doori. 2F = focus se dugni doori ka point.

10.2 RULES FOR RAY DIAGRAMS (LENS)

RULE 1: Axis ke PARALLEL aane wali ray - convex me F se guzarti; concave me F se aati hui lagti.

RULE 2: OPTICAL CENTRE (O) se guzarne wali ray - bina mude SEEDHI nikal jaati.

RULE 3: FOCUS se hokar (convex) / F ki taraf (concave) aane wali ray - refract hokar axis ke PARALLEL ho jaati.

11. IMAGE FORMATION BY LENSES

CONVEX LENS - object positions ke liye image (table yaad rakho):

1) Object at infinity -> Image: at F2; highly diminished; real, inverted
2) Object beyond 2F1 -> Image: between F2 and 2F2; diminished; real, inverted
3) Object at 2F1 -> Image: at 2F2; same size; real, inverted
4) Object between F1 & 2F1 -> Image: beyond 2F2; enlarged; real, inverted
5) Object at F1 -> Image: at infinity; highly enlarged; real, inverted
6) Object between F1 and O -> Image: same side; enlarged; VIRTUAL, erect

CONCAVE LENS - object kahin bhi rakho:

- Image hamesha VIRTUAL, ERECT aur DIMINISHED (chhoti).
- Image object wali taraf, F aur O ke beech banti.

12. LENS FORMULA, MAGNIFICATION AND POWER

LENS FORMULA: 1/v - 1/u = 1/f

(dhyaan do - mirror me PLUS tha, lens me MINUS hai.)

MAGNIFICATION (m): m = v/u = h'/h

- m +ve -> virtual, erect image; m -ve -> real, inverted image.
- |m| > 1 enlarged, |m| < 1 diminished.

POWER OF LENS: P = 1/f (f METRE me, unit = DIOPTRE, D)

- Convex lens ki power +ve; Concave lens ki power -ve.
- Lens jitna zyada mudata (chhota f), utni zyada power.
- Sampark me rakhe lenses ki net power: P = P1 + P2 + P3 + ...

13. SOLVED EXAMPLES (HARDEST -> EASIEST)

Ye type ke questions exam me pakka aate - steps + sign convention khud likhke practice karo.

EXAMPLE 1 (Hardest): Ek CONCAVE mirror ki focal length 10 cm hai. Object usse 15 cm aage rakha hai. Image distance, magnification aur nature batao.

Sign convention: u = -15 cm, f = -10 cm.

Mirror formula: 1/v + 1/u = 1/f

1/v = 1/f - 1/u = (1/-10) - (1/-15) = -1/10 + 1/15

1/v = (-3 + 2)/30 = -1/30 -> v = -30 cm

m = -v/u = -(-30)/(-15) = -30/15 = -2

NATURE: v -ve -> image mirror ke saamne (real). m = -2 (negative aur |m|>1)

-> image REAL, INVERTED aur 2 guna ENLARGED, 30 cm aage banti.

EXAMPLE 2: CONVEX LENS ki focal length 10 cm. Object 20 cm door rakha hai. Image distance aur magnification nikaalo.

Sign convention: u = -20 cm, f = +10 cm.

Lens formula: 1/v - 1/u = 1/f

1/v = 1/f + 1/u = 1/10 + (1/-20) = 1/10 - 1/20 = (2 - 1)/20 = 1/20

v = +20 cm

m = v/u = 20/(-20) = -1

NATURE: v +ve -> doosri taraf REAL image; m = -1 -> inverted aur SAME SIZE.

(Object 2F par tha, isliye image bhi 2F par same size - table se match.)

EXAMPLE 3: Kisi kaanch ka refractive index 1.5 hai. Light ki speed kaanch me nikaalo (c = 3 x 10^8 m/s).

n = c/v -> v = c/n = (3 x 10^8)/1.5 = 2 x 10^8 m/s.

(Snell's form: n = sin i / sin r se bhi i ya r nikaal sakte ho.)

EXAMPLE 4: (a) Ek convex lens ki focal length 25 cm hai - power nikaalo.

(b) Phir P1 = +5 D aur P2 = -2 D ko jodo (contact me) - net power + f.

(a) f = 25 cm = 0.25 m; P = 1/f = 1/0.25 = +4 D.

(b) P = P1 + P2 = (+5) + (-2) = +3 D. Net f = 1/P = 1/3 m = 0.33 m (33.3 cm).

EXAMPLE 5: Concave mirror ke saamne object F aur P (pole) ke BEECH rakha hai - image ki nature, size aur position batao (table se).

Answer: Image VIRTUAL, ERECT aur ENLARGED (badi), mirror ke PEECHE banti.

(Yahi reason shaving mirror chehra paas rakhne par bada seedha dikhata hai.)

EXAMPLE 6: Convex lens ke saamne object 2F se BAHAR (beyond 2F1) rakha hai - ray diagram describe karo aur image batao.

- Ray 1: object ke top se axis ke PARALLEL chalti -> lens ke baad F2 se guzarti.
- Ray 2: object ke top se OPTICAL CENTRE (O) se seedhi (bina mude) nikalti.
- Dono jahan milti wahan image: F2 aur 2F2 ke BEECH; REAL, INVERTED, DIMINISHED.

EXAMPLE 7 (Easiest): Reflection ke do niyam (laws of reflection) likho.

(i) Angle of incidence = angle of reflection (angle i = angle r).

(ii) Incident ray, reflected ray aur normal - teeno point of incidence par EK HI PLANE me hote hain.

14. QUICK Q&A - PADHNE KE BAAD KHUD CHECK KARO

Pehle khud answer socho, fir niche milao. Ye board-style questions hain.

Q1. Real aur Virtual image me farak?
A1. Real = rays sach me milti, screen par aati, inverted (concave mirror/convex lens se). Virtual = rays milti nahi (lagti), screen par nahi aati, erect (plane mirror).

Q2. Concave mirror ka use search-light/headlight me kyun?
A2. Bulb ko F par rakhne se reflect hokar strong PARALLEL beam milti (door tak jaati).

Q3. Convex mirror gaadi ke rear-view me kyun lagate?
A3. Hamesha erect + chhoti image deta aur WIDE area (bada field of view) dikhata, isliye peeche ka zyada traffic dikh jaata.

Q4. Mirror formula aur Lens formula likho.
A4. Mirror: $1/v + 1/u = 1/f$. Lens: $1/v - 1/u = 1/f$.

Q5. Refractive index ka formula aur uska matlab?
A5. $n = c/v$. Batata light medium me kitni dheemi padti; n zyada = optically denser.

Q6. Power of lens kya hai aur unit?
A6. $P = 1/f$ (f metre me). Unit = DIOPTRE (D). Convex +ve, Concave -ve.

Q7. Paani me rakhi pencil tedhi kyun dikhti?
A7. Refraction ki wajah se - paani se aati light surface par mudti, isliye doobaa hissa uthaa/muda dikhta.

Q8. Glass slab se nikalne wali emergent ray ke baare me ek baat?
A8. Wo incident ray ke PARALLEL hoti (bas thodi side khisak jaati = lateral shift).

Q9. Magnification ka sign kya batata?
A9. +ve -> virtual + erect image; -ve -> real + inverted image. $|m|>1$ badi, $|m|<1$ chhoti.

Q10. $f = R/2$ me f aur R kya hain?
A10. f = focal length (P se F), R = radius of curvature (P se C). Focal length, R ka aadha.

EXAM STRATEGY (1 minute revision):
- Laws of reflection + laws of refraction (Snell) zaroor likhna aana chahiye.
- Mirror formula $1/v+1/u=1/f$ aur Lens formula $1/v-1/u=1/f$ (PLUS vs MINUS dhyaan).
- Magnification $m = -v/u$ (mirror), $m = v/u$ (lens) - sign ka matlab.
- Concave mirror image TABLE (6 positions) + convex mirror "hamesha virtual" yaad.
- Sign convention, refractive index $n=c/v$, Power $P=1/f$ (dioptre) - scoring topics.
- Ray-diagram ke 3 rules (mirror aur lens dono) practice karo.
- CORE BASICS page (ray, normal, real/virtual, concave/convex) bhool jaao to wapas pehle padho - base strong rakho.

Generated by Kiro for Krishna1k - All the best!